CAPACITORS

Q 1. A sine wave voltage is applied across a capacitor. When the frequency of the voltage is decreased, the current

(A) Increases

(B) Remains constant

(C) Decreases

(D) Ceases

Q 2. Two series capacitors (one 2 μF, the other of unknown value) are charged from a 24 V source. The 2 μF capacitor is charged to 16 V and the other to 8 V. The value of the unknown capacitor is

(A) 1 μF

(B) 2 μF

(C) 4 μF

(D) 8 Mf

Q 3. Total parallel capacitance is less than that of the smallest capacitor in parallel.

(A) True

(B) False

Q 4. A 2 μF, a 4 μF, and a 10 μF capacitor are connected in series. The total capacitance is less than

(A) 2 μF

(B) 4 μF

(C) 10 μF

(D) 1.5 μF

Q 5. XC is inversely proportional to frequency and capacitance.

(A) True

(B) False

Q 6. An uncharged capacitor and a 1 kΩ resistor are connected in series with a switch and a 6 V battery. At the instant the switch is closed, the voltage across the capacitor is

(A) 6 V
(B) 3 V
(C) 2 V
(D) 0 V

Q 7. A capacitor blocks A.C. but passes D.C.
(A) True
(B) False

Q 8. A 220 Ω resistor is in series with a 2.2 μF capacitor. The time constant is
(A) 48 μs
(B) 480 μs
(C) 2.42 μs
(D) 24 μs

Q 9. The dielectric constant is the ability of a material to establish an electric field.
(A) True
(B) False

Q 10. A capacitor of 0.02 μF is larger than
(A) 0.000020 F
(B) 200,000 pF
(C) 2,000 pF
(D) All of the above

Q 11. A 4.7 MΩ resistor is in series with a 0.015 μF capacitor. The combination is across a 12 V source. How long does it take the capacitor to fully charge?
(A) 35 ms
(B) 352 ms
(C) 3.5 s
(D) 70.5 ms

Q 12. Two 0.68 μF capacitors are connected in series across a 10 kHz sine wave signal source. The total capacitive reactance is

(A) 46.8 Ω
(B) 4.68 Ω
(C) 23.41 Ω
(D) 11.70 Ω

Q 13. Capacitance is directly proportional to plate area and plate separation.
(A) True
(B) False

Q 14. A 10 μF, 20 μF, 22 μF, and 100 μF capacitor are in parallel. The total capacitance is
(A) 2.43 μF
(B) 4.86 μF
(C) 100 μF
(D) 152 μF

Q 15. Voltage leads current by 90° in a capacitor.
(A) True
(B) False

Q 16. A capacitor of 2,000 pF is greater than
(A) 0.02 μF
(B) 0.002 μF
(C) 0.0002 μF
(D) 0.0000002 F

Q 17. The true power in a capacitor is zero.
(A) True
(B) False

Q 18. A 12 kHz voltage is applied to a 0.33 μF capacitor, and 200 mA of rms current is measured. What is the value of the voltage?

(A) 8 V
(B) 80 V

(C) 80 mV
(D) 800 mV

Q 19. The ohm is the unit of capacitance.
(A) True
(B) False

Q 20. When the plate area of a capacitor increases,
(A) The capacitance increases
(B) The capacitance decreases
(C) The capacitance is unaffected
(D) The voltage it can withstand increases

Q 21. When the voltage across a capacitor is tripled, the stored charge
(A) Triples
(B) Is cut to one-third
(C) Stays the same
(D) Doubles

Q 22. A 0.47 µF capacitor is across a 2 kHz sine wave signal source. The capacitive reactance is
(A) 170 Ω
(B) 17 Ω
(C) 0.000169 Ω
(D) 1.7 Ω

Q 23. Four 0.15 µF capacitors are in parallel. The total capacitance is
(A) 0.15 µF
(B) 0.30 µF
(C) 0.6 µF
(D) 0.8 µF

Q 24. Which of the following capacitors is polarized?
(A) Mica
(B) Ceramic

(C) Plastic-film
(D) Electrolytic

Answer: Option D
Q 25. Energy is stored by a capacitor in a magnetic field.
(A) True
(B) False

Q 26. Of the following capacitors, which one has the highest dielectric constant?
(A) Air
(B) Mica
(C) Glass
(D) Paper

Q 27. A 0.01 µF capacitor equals how many picofarads?
(A) 100 pF
(B) 1,000 pF
(C) 10,000 pF
(D) 100,000 pF

Q 28. A capacitor and a resistor are connected in series to a sine wave generator. The frequency is set so that the capacitive reactance is equal to the resistance and, thus, an equal amount of voltage appears across each component. If the frequency is increased
(A) VR > VC
(B) VC > VR
(C) VR = VC
(D) VR and VC = 0

Q 29. A 0.00022 F capacitor equals how many microfarads?
(A) 22 µF
(B) 220 µF
(C) 2,200 µF
(D) 22,200 µF

Q 30. A 330 pF capacitor and a 220 pF capacitor are each connected across a 6 V dc source. The voltage across the 330 pF capacitor is

(A) 3 V

(B) 6 V

(C) 4 V

(D) 0 V

Q 31. Five time constants are required for a capacitor to charge fully or discharge fully.

(A) True

(B) False

Q 32. What is the capacitance when Q = 60 μC and V = 12 V?

(A) 720 μF

(B) 5 μF

(C) 50 μF

(D) 12 μF